Collective Cognition in the Age of AI

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Proposed format: upper-level undergraduate seminar; 14 teaching weeks; two 75-minute meetings per week. Intended for students across the humanities and social sciences, as well as computer science.

Course description

Collective cognition, the process through which groups acquire, process, and strategically leverage information, is central to humans' exceptional ability to solve complex problems and scaffold increasingly sophisticated culture. Recently, artificial intelligence has acquired significant potential to dramatically alter these abilities, for better or for worse. To address this uncertainty about the long-term implications of AI, the course revisits key themes in collective cognition in light of recent research on human-AI interactions. Each week, foundational papers from diverse perspectives on collective cognition (e.g., cultural evolution, philosophy of science, and political science) are discussed in parallel with recent AI research.

The course has three goals. The first aim is to familiarize students with the notion of collective cognition and its relevance to social, political, and cultural aspects of contemporary society. The second aim is to provide students with intellectual tools to navigate significant societal transformations as AI takes an increasingly large place in our society. Finally, the third aim is to help students design collaborative research projects.

Learning outcomes

By the end of the course, students will be able to:

- Understand the dynamics that can improve or challenge the effectiveness of decentralized collectives.
- Link challenges and opportunities arising from AI to fundamental issues in collective cognition.
- Collaborate on a large-scale project with important coordination needs.
- Design experimental research on human-AI interactions.

Teaching approach

The course combines reading based discussions (22.5 hours), hands-on tutorials (5 hours), and a joint research project (7.5 hours).

Students will be invited to transform the classroom itself into an experiment by self-organizing around a joint research project on AI in collective cognition. They will elaborate a research question, a methodology, a division of labor, and submit a collective report. They will reflect on the process they adopted and how their struggles connect to more fundamental issues in collective cognition.

There will be three kinds of sessions:

- **Readings (22.5 hours):** The instructor introduces the theme of the week and the main readings. Each session, one to two foundational papers are discussed in parallel with one to two contemporary AI papers illuminating how core issues in collective cognition may be disrupted by AI. Students read one paper of each kind per week (or more if they want to). According to

a predetermined schedule, each paper is presented by one student, who must briefly describe how it could inform the group project. A discussion follows to explore the connections between the readings and the group project.
- **Practical sessions (5 hours):** The instructor gives a hands-on tutorial on methodological tools (based on PsyNet) that the students can use to design their joint research project.
- **Group project (7.5 hours):** Students work together on their project in-class with the instructor's guidance and help. Students should coordinate and work outside the class and use this time to get feedback from the instructor.

All sessions end with a brief discussion on the status of the group project.

Prerequisites

Students should come from diverse disciplinary backgrounds, as diversity benefits the group project. However, students should have at least one of the following: training in philosophy or social sciences; training in computer science; training in experimental methods in a relevant area (psychology, behavioral economics, etc.).

Assessment

- **Engagement with the readings (25%).** Three unannounced in-class written quizzes (15 minutes each) will be given to assess the students' understanding of the readings.
- **Oral presentations (20%).** Students will be assigned at least one reading to present in class in three minutes + Q&A, focusing on how the reading could inform the group project.
- **Group project (40%).** The grade comes from the final joint product (25%) and individual reports (75%) Students describe their own contribution to the group project in an individual report.
- **Individual reflection on the project (15%).** In their individual report, students will also reflect critically on challenges met during the project, and how they connect with fundamental issues in collective cognition such as coordination problems, free-riding, and AI use.

AI policy

Students are free to use AI tools with transparent and accurate attribution. For the joint research project, students will be encouraged to agree on a policy of their own, as long as it is at least as strict as the classroom policy.

Timeline

1. From individual minds to collective cognition (weeks 1 and 2)

Guiding question. What enables collective cognition at a societal level?

Foundational papers

- Hayek (1945). The Use of Knowledge in Society. *American Economic Review*.
- Ober (2008). Introduction: Dispersed Knowledge and Public Action. *Democracy and Knowledge: Innovation and Learning in Classical Athens*, pp. 1–38.
- Muthukrishna and Henrich (2016). Innovation in the collective brain. *Philosophical Transactions of the Royal Society B: Biological Sciences*.

Contemporary AI papers

- Farrell et al. (2025). Large AI models are cultural and social technologies. *Science*.
- Burton et al. (2024). How large language models can reshape collective intelligence. *Nature Human Behaviour*.

2. The wisdom and madness of crowds (weeks 3 and 4)

Guiding question. When are collectives smarter together, and when are they dumber?

Foundational papers

- Hong and Scott E. Page (2004). Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*.
- Lorenz et al. (2011). How social influence can undermine the wisdom of crowd effect. *Proceedings of the National Academy of Sciences*.
- Bikhchandani et al. (1992). A Theory of Fads, Fashion, Custom, and Cultural Change as Informational Cascades. *Journal of Political Economy*.

Contemporary AI papers

- Vaccaro et al. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*.
- Berger et al. (2026). Beyond AI advice – independent aggregation boosts human-AI accuracy. arXiv preprint arXiv:2603.29866.
- Acemoglu et al. (2026). How AI Aggregation Affects Knowledge. NBER Working Paper 35036.

3. Exploration versus exploitation and incentive structures in collective cognition (weeks 5 and 6)

Guiding question. Who bears the costs of learning?

Week 5: Exploration and exploitation

Foundational readings

- March (1991). Exploration and Exploitation in Organizational Learning. *Organization Science*.
- Centola (2022). The network science of collective intelligence. *Trends in Cognitive Sciences*.

Contemporary AI reading

- Li et al. (2026). Human-AI Synergy Supports Collective Creative Search. arXiv preprint arXiv:2602.10001.

Week 6: Incentives and credit

Foundational readings

- Kummerfeld and Zollman (2016). Conservatism and the Scientific State of Nature. *The British Journal for the Philosophy of Science*.
- Šešelja (2023). Agent-Based Modeling in the Philosophy of Science. *The Stanford Encyclopedia of Philosophy*. Especially §3.4, "Evolutionary game-theoretic models of bargaining."
- Deffner et al. (2024). Collective incentives reduce over-exploitation of social information in unconstrained human groups. *Nature Communications*.

Contemporary AI readings

- Jo and Raghavan (2026). Incentives Shape How Humans Co-Create with Generative AI. arXiv preprint arXiv:2604.03529.
- Rio-Chanona et al. (2024). Large language models reduce public knowledge sharing on online Q&A platforms. *PNAS Nexus*.
- Castellacci et al. (2026). Generative Artificial Intelligence in Scientific Research: Individual Benefits, Collective Risks, and a Framework for Responsible Research with AI. arXiv preprint arXiv:2607.24879.

4. Methodology & Group Project

Guiding question. How can we study hybrid human-AI societies experimentally?

Week 7: Human-AI study design and modeling

- Lai et al. (2023). Towards a Science of Human-AI Decision Making: An Overview of Design Space in Empirical Human-Subject Studies. *2023 ACM Conference on Fairness Accountability and Transparency*.
- Xu et al. (2026). Large language models for computational social science. *Nature Human Behaviour*.
- Sucholutsky et al. (2025). Using LLMs to advance the cognitive science of collectives. *Nature Computational Science*.

Week 8: Designing human-AI experiments (practical sessions)

Using an instructor-provided PsyNet template, students translate a provided research question into an experimental design, specifying the human-AI interaction, comparison conditions, and outcome measures.

Week 9: Designing human-AI experiments (practical sessions)

Students test and refine their experiment in a classroom pilot and inspect the resulting data.

5. The resilience of cumulative cultural evolution (week 10)

Guiding question. Can we accumulate knowledge without understanding?

Foundational readings

- Henrich (2015). Lost European Explorers. *The Secret of Our Success: How Culture Is Driving Human Evolution, Domesticating Our Species, and Making Us Smarter*, pp. 22–33.
- Henrich (2004). Demography and Cultural Evolution: How Adaptive Cultural Processes Can Produce Maladaptive Losses—The Tasmanian Case. *American Antiquity*.
- Derex et al. (2019). Causal understanding is not necessary for the improvement of culturally evolving technology. *Nature Human Behaviour*.

Contemporary AI papers

- Messeri and Crockett (2024). Artificial intelligence and illusions of understanding in scientific research. *Nature*.
- Aslanov et al. (2025). Overconfidence without Understanding: AI Explanations Increase the Illusion of Explanatory Depth. PsyArXiv preprint.

6. Democratic deliberation (week 11)

Guiding question. Does deliberation produce new knowledge?

Foundational reading

- Landemore (2012). Democratic Reason: The Mechanisms of Collective Intelligence in Politics. *Collective Wisdom*, pp. 251–289. Read the linked version through Section 3 included.

Contemporary AI papers

- Tessler et al. (2024). AI can help humans find common ground in democratic deliberation. *Science*.
- Salvi et al. (2025). On the conversational persuasiveness of GPT-4. *Nature Human Behaviour*.
- Gaubert et al. (2026). An Experimental Method to Study Opinion Diffusion in Human-AI Hybrid Societies. *Proceedings of the Annual Meeting of the Cognitive Science Society*.

Weeks 12 to 14: shared project and synthesis

Students use the classroom to work on the group project collaboratively while receiving direct feedback from the instructor. The instructor gives short lectures and focused tutorials based on the needs of the class. On the last week, students present the final synthesis and receive feedback to help them write their individual reports.

Helpful readings

- Zollman (2007). The Communication Structure of Epistemic Communities. *Philosophy of Science*.
- Zollman (2010). The Epistemic Benefit of Transient Diversity. *Erkenntnis*.
- Shin et al. (2023). Superhuman artificial intelligence can improve human decision-making by increasing novelty. *Proceedings of the National Academy of Sciences*.
- Chromik et al. (2021). I Think I Get Your Point, AI! The Illusion of Explanatory Depth in Explainable AI. *26th International Conference on Intelligent User Interfaces*.
- Bastani et al. (2025). Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*.
- Budzyń et al. (2025). Endoscopist deskilling risk after exposure to artificial intelligence in colonoscopy: a multicentre, observational study. *The Lancet Gastroenterology & Hepatology*.
- Read (2006). Tasmanian Knowledge and Skill: Maladaptive Imitation or Adequate Technology? *American Antiquity*.
- Doshi and Hauser (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances*.
- Shiiku et al. (2026). Hybrid Human-AI Societies Balance Creativity and Diversity. PsyArXiv preprint.
- Scott E Page (2025). Everyone, everywhere, all at once LLMs and the new physics of collective intelligence. *Collective Intelligence*.
- Shumailov et al. (2024). AI models collapse when trained on recursively generated data. *Nature*.
- Wilkens (2026). AI as a Tool for Simulation-Based Experiments in Literary Studies. arXiv preprint arXiv:2606.02293.
- Guilbeault et al. (2026). Artificial Intelligence and the New Science of Culture. arXiv preprint

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